

Alejandro Sánchez Yalí

Product Engineer



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SUMMARY

Product Engineer who takes products from an idea to a deployed system. In the last two years led three complete products – Plixiq (AI customer support on WhatsApp), Aluna (AI-assisted recruitment) and VitaStock (surgical supply chain for private clinics) – owning discovery with users, architecture, backend, frontend and deploy. 8+ years of engineering, an M.Sc. in Mathematics and 11 years teaching, which is why the hardest parts of the job – explaining a system and deciding what not to build – come naturally.

TECHNICAL SKILLS

- **Ship the whole thing:** Python, FastAPI, Django, Celery, TypeScript, Next.js, React, Node.js, NestJS, Rust (Axum), PostgreSQL, Redis, Docker, Railway, AWS
- **AI in products:** LLMs, LangChain, LangGraph, LiteLLM, RAG, MCP, OpenAI API, Whisper, ElevenLabs, WhatsApp Cloud API
- **Decide well:** Clean architecture, DDD, SOLID, modular monolith, event-driven design, trade-off analysis, technical design sessions
- **Keep it alive:** GitHub Actions, CI/CD, Docker, background jobs, real-time dashboards, TDD, Vitest, Jest, code review

PRODUCTS SHIPPED

- **Plixiq – AI Customer Support on WhatsApp** [↗](#) – *Python, FastAPI, LiteLLM, PostgreSQL, Next.js, Railway*
 - Took the product from an idea to a deployed system as architect and lead engineer: discovery, architecture, backend, frontend and deploy
 - Near-complete MVP in 4 months, covered by about 590 backend tests and CI on every pull request; infrastructure runs at \$60–80/month because cost scales with LLM usage
- **Aluna – AI Recruitment Platform for Staffing Agencies** [↗](#) – *Next.js 15, FastAPI, LiteLLM, PostgreSQL, Inngest, Claude, Gemini, WhatsApp Cloud API*
 - Lead engineer of an AI recruitment platform for staffing agencies: CVs are scored automatically, an agent interviews candidates over WhatsApp or web chat, and the recruiter gets a shortlist and a client-ready report
 - Deliberate model routing through LiteLLM (Gemini to read documents, Claude to converse), multi-tenant roles from a single policy source, and Ley 1581/2012 consent, deletion receipts and audit logs built into the data model
- **VitaStock – Surgical Supply Chain for Clinics** [↗](#) – *Python, FastAPI, Next.js, TypeScript, PostgreSQL, Docker, Railway*
 - Architect and lead engineer of a platform commissioned by a private surgical clinic in Colombia – 12 modules and 5 user roles designed to replace spreadsheet-based inventory and procurement; delivered, with rollout at the clinic pending
 - Surgery scheduling with supply baskets and real-time consumption tracking, lot-level traceability with expiry alerts, and true per-patient cost through a cardex

EXPERIENCE

Independent AI Consultant
Freelance

Oct 2026 – Present
Remote

- Lead engineer of Aluna, an AI recruitment platform for staffing agencies (see below)
- Use Claude Code daily on live client projects, reviewing every generated change like a pull request before it ships

AI Specialist
Lapzo

Aug 2025 – Sep 2026
Remote, Mexico

- Designed and built a Digital Professor that converses with learners in real time and teaches through interactive slides
- Made the key architectural calls: which LLM orchestration stack, how voice synthesis plugs into the conversation, where the service boundaries go

Independent Software & AI Consultant

Freelance

Aug 2024 – Jul 2025

Remote

- Lead engineer of CREARIA’s multi-tenant AI support agent on WhatsApp, email and Instagram, in production with several client businesses including a real-estate agency: led a team of three across 5 FastAPI services and 2 Next.js dashboards on AWS ECS with Terraform, RAG over pgvector, 11 MCP tools scoped per tenant, LiteLLM fallback routing and human escalation; load-tested to 250 virtual users. Also designed the architecture of an AI coaching agent for one of CREARIA’s clients
- Ran technical design sessions with clients, turning vague business problems into architectures and naming the trade-off behind every technology choice
- Code reviewer for a client’s development team and architect of their workflow automation platform’s database design

Full Stack Engineer

Monadical

Apr 2021 – Aug 2024

Remote, Canada/US

- Three years on a distributed team across Canada, the US and Latin America, shipping full stack products in Python and TypeScript
- Full-stack engineer on ZoHuddle, a video conferencing app for scientific conferences (Neuromatch) that matches participants by mutual interests and supports conferences of about 200 participants: React front end and Django back end, with Django Channels (WebSockets) and Twilio Video
- Developed both frontend and backend, with tests; did code reviews

Full Stack Engineer

BCFort

Aug 2018 – Oct 2020

Medellín, Colombia

- Designed system architecture for blockchain and analytics platforms, from smart contract patterns to service boundaries
- Built decentralised applications and NFT marketplaces with React, Web3.js, Ethereum and Hyperledger

Professor of Mathematics & Physics (part-time)

Universidad de Antioquia

Jan 2010 – Oct 2021

Medellín, Colombia

- Eleven years teaching mathematics and physics, the last year and a half also machine learning, and building the AI/ML elective curriculum for the Mathematics Institute
- Mentored student research projects – the same work that later became code review and design docs

OPEN SOURCE

- [talking-avatar-with-ai](#)  – 464 stars, 110 forks. Open-source conversational AI avatar: OpenAI GPT, Whisper, ElevenLabs, Rhubarb lip-sync, Three.js
- [Monadical-SAS/Morpheus](#)  – 19 pull requests. Open-source AI image generation platform: FastAPI, React, Stable Diffusion
- [kipux](#)  – Collaborative-finance app in Rust (Axum, sqlx, PostgreSQL, HTMX, SSE, PWA) with Claude-vision receipt parsing. Open source (MIT), work in progress

TALKS & WRITING

- [Scaling Learning Models: Parallelism Strategies in JAX with Flax](#)  – PyCon Colombia, 2024
- **Engineering case studies:** [Plixiq](#)  · [CREARIA Agent](#)  · [Aluna](#)  – architecture and trade-offs behind each system (2026)
- [Revolutionizing Animation: Building Digital Humans with LLMs](#) (2024)

EDUCATION

M.Sc. in Mathematics

Universidad de Antioquia

Aug 2010 - Sep 2013

Medellín, Colombia

B.Sc. in Teaching Mathematics and Physics

Universidad de Antioquia

Jan 2004 - Oct 2009

Medellín, Colombia